

Master’s Thesis Exposé

An Evidence-Generation Framework for Digital Therapeutics (DTx): An LLM-driven Approach to Integrating Clinical, Observational, and Financial Evidences of DTx

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Abstract

Digital Therapeutics (DTx), in Germany known as *DiGA* (Digitale Gesundheitsanwendungen), operate in a complex regulatory environment requiring specific evidence for reimbursement. While Randomized Controlled Trials (RCTs) remain the regulatory *Gold Standard*, Real-World Evidence (RWE) is increasingly vital for demonstrating market fit and long-term efficacy. This thesis proposes a scalable, semi-automated framework utilizing Large Language Models (LLMs) to extract, classify, and link evidence (both RCT and RWE) with financial business data for DTx. By benchmarking LLM performance against a manually annotated *label test set* of 73 German DiGAs and validating generalizability on a US market subset, this research aims to provide a robust *Data Layer* that supports DTx evidence analysis.

1 Introduction and Motivation

Germany’s Digital Healthcare Act (DVG) of 2019 introduced a *Fast-Track* process for the reimbursement of digital health applications (DiGA) [1,2]. Manufacturers must demonstrate a *positive healthcare effect* (pVE) through comparative studies. Traditionally, this follows a sequential model: expensive Randomized Controlled Trials (RCTs) are conducted for regulatory approval, potentially, followed by post-market Real-World Evidence (RWE) collection [2].

However, this sequential approach may reduce/hinder innovation and market viability. Investors and stakeholders increasingly demand a *parallel approach*, where regulatory efficacy (via RCTs) and market fit (via RWE/observational data) are established simultaneously. Analyzing this landscape is currently hindered by the unstructured nature of evidence data, which is scattered across registries (e.g., ClinicalTrials.gov, DRKS.de), publications (e.g., PubMed), company websites, and capital market data (e.g., Pitchbook).

Recent advancements in LLMs offer a potential solution for automating the extraction and classification of complex evidence generation strategies from unstructured text [3]. This thesis aims to provide a specific contribution by developing an automated pipeline

to structure this evidence ecosystem, enabling a data-driven analysis of whether parallel evidence generation correlates with financial success.

2 Research Questions

The central research question (RQ) is:

RQ: How can an LLM-based framework be designed and validated to collect, link, and structure clinical trials (RCT), observational evidence (RWE), and financial data for the DTx ecosystem?

To address this, the following sub-questions will be investigated:

- **Sub-RQ 1 (Taxonomy & Schema):** What is the optimal classification schema to distinguish and categorize heterogeneous evidence sources (RCT and RWE) for DiGAs?
- **Sub-RQ 2 (LLM Performance & Model Selection):** How does the extraction performance of state-of-the-art commercial models (e.g., GPT-5, Gemini 3) compare to open-source alternatives (e.g., Llama 4, Mistral) when benchmarked against a manual *Gold Standard*?
- **Sub-RQ 3 (Generalization):** Can an LLM-based framework trained/prompted on German regulatory data be successfully transferred to the less regulated US digital health market to extract comparable evidence profiles?

3 Methodology

The project adopts a quantitative, engineering-focused approach involving web scraping, Natural Language Processing (NLP), and comparative evaluation.

3.1 Data Collection and Sources

Data will be aggregated from three primary domains:

1. **Company Data:** Official websites of DTx (via scraping) to identify stated evidence from the companies themselves and non-indexed whitepapers.
2. **Regulatory & Clinical Data:** The BfArM DiGA directory (approx. 73 approved DiGAs), DRKS (German Clinical Trials Register), ClinicalTrials.gov, and PubMed.
3. **Financial Data:** Funding rounds and investor data linked via Pitchbook.com or Crunchbase.com APIs or scraping.

3.2 Proposed Framework

The pipeline will consist of three stages:

- **Scraping Module:** A semi-automated browser automation pipeline using different providers (e.g., Browser-Use, Apify) to retrieve relevant text, links, and PDFs.
- **LLM Extraction Layer:** Utilization and benchmarking of both **commercial** (via API) and **open-source** (local/hosted) state-of-the-art LLMs with Retrieval-Augmented Generation (RAG) to populate the classification schema.
- **Data Linkage:** Automated matching of clinical evidence to corporate financial profiles, including investment rounds and their sizes.

3.3 Validation Strategy

Validation is a critical component of this thesis:

- **Label Test Set (Germany):** The extraction performance of the LLM models (commercial vs. open-source) will be benchmarked against a manually created *Gold Standard* dataset of the 73 German DiGAs. Comparative metrics such as Precision, Recall, and F1-score will be calculated for key fields.
- **Transferability Test (USA):** To demonstrate generalization, the framework will be applied to a subset of approximately 50 US-based DTx companies. This tests the system’s robustness outside the specific German regulatory context.

Work Plan and Timeline

The thesis is scheduled for 6 months, starting in December 2025.

Table 1: Master Thesis Work Plan and Timeline (6 Months)

Month	Timeframe	Key Tasks	Milestones
1	Dec 2025 – Jan 2026	• Literature review and synthesis. • Refinement of the extraction taxonomy. • Setup of the scraping environment.	
2	Feb 2026	• Development of the LLM extraction pipeline. • Implementation of dual-inference logic (Commercial API vs. Open-Source Local). • Prompt engineering and initial testing.	
3	Mar 2026	• Execution on the German label test set. • Comparative benchmarking of candidate models against the manual database.	Presentation 1: German Label Set Results
4	Apr 2026	• Iteration and optimization of the pipeline. • Expansion to the US market dataset (approx. 50 companies).	
5	May 2026	• Analysis of results (generalization to US data). • Linking and harmonization of datasets.	Presentation 2: Generalization of US Data
6	Jun 2026	• Comprehensive writing of the thesis. • Final revisions and preparation for defense.	Final Submission and Presentation

References

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- [3] M. Agrawal *et al.*, “Large language models are few-shot clinical information extractors,” in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024. [Online]. Available: <https://arxiv.org/abs/2205.12689>